RATING METHODOLOGY

10 October 2025

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Rating Methodology

Automobile Manufacturers

This rating methodology replaces the *Automobile Manufacturers* methodology published on 11 April 2025. We have made editorial changes throughout the methodology to enhance readability. These updates do not change our methodological approach.

Scope

This rating methodology applies to companies globally that are primarily engaged in the design and manufacture of passenger vehicles, including cars, light trucks and motorcycles.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

We rate companies that are primarily engaged in the production and sale of other types of vehicles or related products, including large commercial vehicles, automobile components and other industrial products, using other methodologies, such as our methodologies for automotive suppliers and manufacturing companies.[1]

The universe of companies that we rate using this methodology also excludes the captive finance subsidiaries of automobile manufacturers, which we rate using our methodology for captive finance subsidiaries of nonfinancial corporations. Captive finance subsidiaries provide automotive financing services on the wholesale level, i.e., sales from the automobile manufacturer to the dealer network, and the retail level, i.e., sales from dealers to consumers.

Sector overview

Automobile manufacturing is a global industry, with most companies operating across multiple continents. Companies with stronger credit profiles typically are larger and have a global footprint, with automotive operations in the key markets of North America, Europe, Japan and China. They typically have a strong market position and competitive products across a broad range of vehicle categories. Companies with weaker credit profiles tend to be smaller and less established in key regional markets. Also, they typically have more limited product diversity and lower profitability.

High cash flow volatility is a characteristic of automobile manufacturers, owing to the cyclical nature of the industry and high degree of competition within each market. Many companies operate with materially negative working capital, driven by late payments to suppliers. Some companies maintain high cash balances to manage operating cash flow volatility as well as unexpected funding needs of captive finance activities. Economic downturns, as well as company-specific circumstances that consume cash, make financial flexibility important. Such circumstances may include the potential for high product development costs, product defects or recalls, or capital calls from a captive finance subsidiary.

Consumer sentiment tends to follow macroeconomic, social and product trends, requiring companies to actively manage their product development cycles to stay competitive. This results in high capital intensity, with companies typically engaged in robust and ongoing reinvestment.

Because of its fundamental role in the global economy, the automobile industry is the subject of significant political and regulatory attention across jurisdictions. Political and regulatory intervention can take many forms and can be negative or positive for a company's credit profile. Government scrutiny of environmental and safety standards can create additional costs, but financial support in times of stress has proved critical.

The information in this methodology applies to automobile manufacturing company standalone financial statements and excludes the contribution from captive finance operations where publicly available financial information permits. For a discussion of the impact of captive finance subsidiaries on the standalone ratings of parent automobile manufacturers, see the "Other considerations" section.

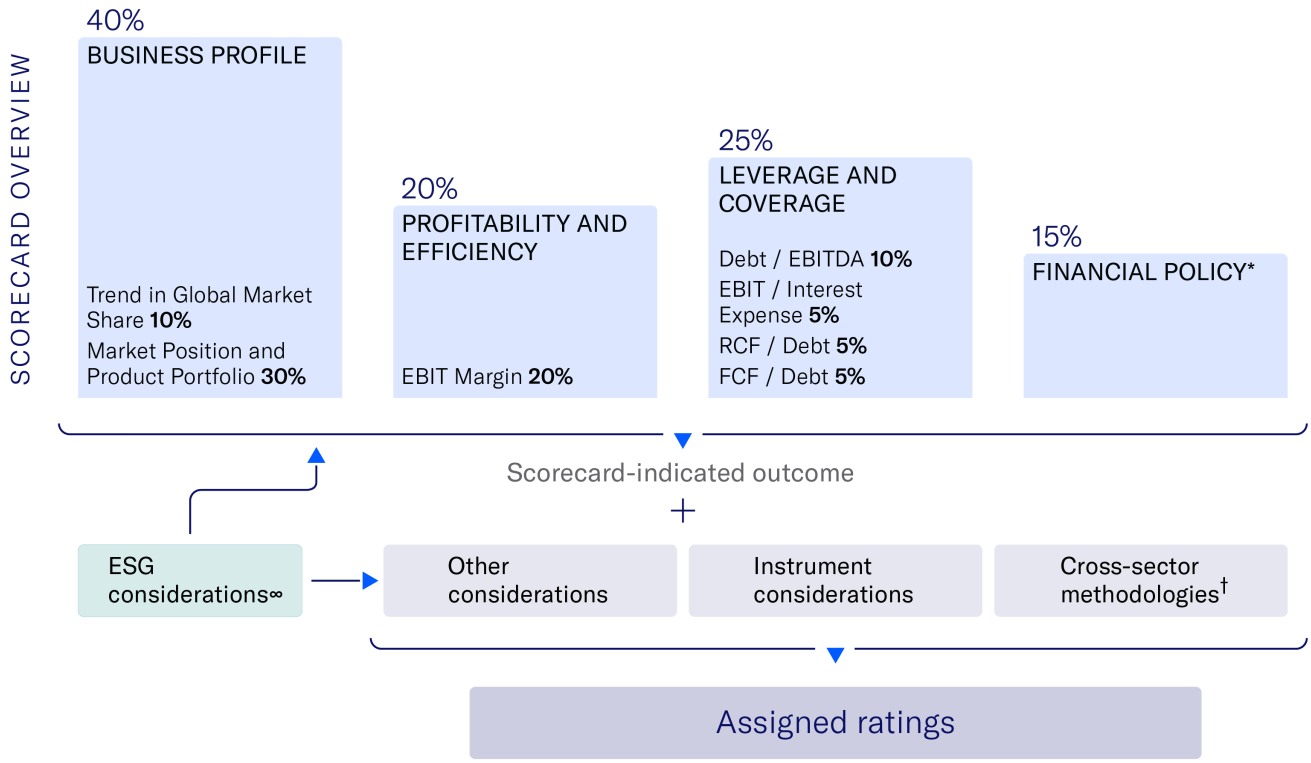
Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1

Illustration of the automobile manufacturers rating approach



* This factor has no subfactors.

∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.

† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Automobile manufacturers scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Business Profile (40%)									
Trend in Global Market Share[1]	10%	Increase > 2%	Increase >1% but ≤ 2%	Increase > 0.25% but ≤ 1%	Increase ≤ 0.25% to decrease ≤ 0.25%	Decrease > 0.25% but ≤ 2%	Decrease > 2% but ≤ 3%	Decrease > 3% but ≤ 4%	Decrease > 4%
Market Position and Product Portfolio	30%	Very strong competitive position in most of the four key global geographic regions (North America, Europe, Japan and China); very strong and profitable product offering in all segments relevant to those markets,[2] supported by competitive and frequent model launches. Emissions-reducing technologies and AFV[3] product development are extremely strong and clearly lead the market in terms of innovation and customer acceptance; sophisticated, very well-articulated strategy and robust, consistent investments to meet future regulatory standards, including emissions and fuel economy.	Strong competitive position in at least two of the four key global geographic regions; highly competitive and profitable product portfolio in its entirety with a stable rate of model renewal, though there may be some gaps/areas of weakness. Emissions-reducing technologies and AFV product development are very strong with a leading (top quintile) market position in terms of innovation and customer acceptance; comprehensive, well-articulated strategy and mostly ample, consistent investments to meet future regulatory standards, including emissions and fuel economy.	Strong competitive position in at least two of the four key global geographic regions; highly competitive and profitable product portfolio in its entirety with modest variations in rate of model renewal, though there may be some gaps/areas of weakness. Emissions-reducing technologies and AFV product development are strong with an above-average market position in terms of innovation and customer acceptance; well-articulated strategy and consistent investments sufficient to meet future regulatory standards, including emissions and fuel economy.	Strong competitive position with a degree of concentration in one of the four key global geographic regions; product range mostly competitive and profitable with moderate variations in the rate of model renewal, though there may be some gaps/areas of weakness; earnings rely on specific products or segments. Emissions-reducing technologies and AFV product development are sufficient with an average market position in terms of customer acceptance; reasonably well-articulated strategy and generally sufficient investments to meet future regulatory standards for emissions and fuel economy.	Defensible competitive position focused on one of the four key global geographic regions; gaps/areas of weakness in the product portfolio exist with evidence of material variations in the rate of model renewal; earnings rely on specific products or segments or returns on key products cause periodic pressure. Emissions-reducing technologies and AFV product development may be somewhat lagging, but company has articulated a strategy and concrete plans to attain an industry-average market position in terms of customer acceptance; or investments to meet future regulatory standards are uneven, but company has reasonable leeway before delays will have a meaningful impact on ability to meet future regulatory standards.	Weak or eroding competitive position in key geographic regions or product categories with material and chronic variations in product renewal rate or an extended time frame for the renewal of critical models; evidence of weak profitability across most of the model range. Emissions-reducing technologies and AFV product development or investments are lagging, and leeway is limited before delays will have a meaningful impact on ability to meet future regulatory standards.	Long-term viability in one or more key geographic regions or product categories could be threatened, with material and chronic variations in product renewal rate or an extended time frame for the renewal of critical models; product range may not be able to sustain adequate reinvestment needs and significant operational restructuring is required. Extremely limited regulatory flexibility: material challenges to develop (or acquire) and implement emissions-reducing technologies or meet future regulatory standards.	Long-term viability in one or more key geographic regions or in a critical product segment is at clear risk, with structural profit issues and inadequate rate of model renewal. Meaningful current impact from regulation: material challenges or inability to develop (or acquire) and implement emissions reducing technologies are causing a failure to meet regulatory targets.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Profitability and Efficiency (20%)									
EBIT Margin[4]	20%	≥ 20%	15% - 20%	10% - 15%	7% - 10%	4% - 7%	1% - 4%	0.5% - 1%	< 0.5%
Factor: Leverage and Coverage (25%)									
Debt / EBITDA[5]	10%	≤ 0.5x	0.5x - 1.5x	1.5x - 2.5x	2.5x - 3.5x	3.5x - 4.5x	4.5x - 6x	6x - 9x	> 9x
EBIT / Interest Expense[6]	5%	≥ 25x	15x - 25x	10x - 15x	5x - 10x	2x - 5x	1x - 2x	0.5x - 1x	< 0.5x
RCF / Debt[7]	5%	≥ 100%	75% - 100%	50% - 75%	30% - 50%	20% - 30%	10% - 20%	5% - 10%	< 5%
FCF / Debt[8]	5%	≥ 45%	30% - 45%	20% - 30%	10% - 20%	5% - 10%	0% - 5%	(5)% - 0%	< (5)%
Factor: Financial Policy (15%)									
Financial Policy	15%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] This metric is not scored with linear interpolation.
[2] These segments may include luxury, mid-price and economy cars; light trucks and sport utility vehicles; crossover vehicles; and alternative fuel vehicles.
[3] AFV stands for alternative fuel vehicle.
[4] For the linear scoring scale, the Aaa endpoint value is 40%. A value of 40% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[5] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[6] For the linear scoring scale, the Aaa endpoint value is 40x. A value of 40x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5.
[7] For the linear scoring scale, the Aaa endpoint value is 140%. A value of 140% or better equates to a numeric score of 0.5. The Ca endpoint value is 2.5%. A value of 2.5% or worse equates to a numeric score of 20.5.
[8] For the linear scoring scale, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is (10)%. A value of (10)% or worse equates to a numeric score of 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[2] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Business Profile (40% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

Trend in Global Market Share

Market share trends are an important indicator of an automobile manufacturer's ability to sustain its competitive position in its core market segments. Increases or decreases in market share provide insight into an automobile manufacturer's ability to offer, over an average product life cycle, a portfolio of models that remain attractive to consumers across a wide array of vehicle categories.

Declining or volatile market share may indicate that a company's model renewal schedule is uneven, with weak renewal rates in some years compensated only partially by higher renewal rates in other years, which in turn may be an indication that capacity to meet market demand is weak. The introduction of a high-profit, strategically important vehicle can temporarily boost a company's market share position and operating performance. However, as that product ages, share position and performance are likely to weaken until the next reintroduction of the product. Conversely, sustained growth in market share typically reflects a more consistent model renewal rate and a stronger capacity to meet demand in key markets.

Market Position and Product Portfolio

Market position and product portfolio are meaningful indications of how well an automobile manufacturer will be able to withstand changes in market demand. A strong competitive position in the key auto markets of North America, Europe, Japan and China, combined with a profitable product offering, can make a company more resilient to cyclical downturns than a company that relies significantly on specific regions or market subsegments.

The breadth and strength of a company's product portfolio is a critical indicator of its ability to sustain its competitive position. Companies that make continuous investments in new technologies related to vehicle safety, carbon emissions and performance are better able to differentiate their products in the marketplace and avoid consumer perceptions that product features are stale or outdated. Failing to maintain a strong product image could exacerbate the effects of a cyclical downturn on a given company, materially weakening its chances of recovery.

A diverse vehicle lineup is also important, because of the crucial role consumer sentiment plays in automobile demand. Gasoline prices, regulatory mandates and environmental concerns can cause demand to shift toward or away from specific vehicle types. For instance,

sales of large trucks and sport utility vehicles (SUVs) could be at risk in periods of high gasoline prices. If gasoline prices are low and consumers are more focused on safety, larger vehicles may have broader appeal. Consequently, a company with a narrow product range may be more vulnerable to changes in demand patterns than a company that offers a full range of vehicles.

How we assess it for the scorecard

We score this factor based on two quantitative and qualitative subfactors: Trend in Global Market Share; and Market Position and Product Portfolio.

Trend in Global Market Share

We score this subfactor based on our assessment of the quantitative changes in global market share by units (i.e., number of passenger vehicles sold) over the last three years. Where an automobile manufacturer's business focuses on a specific, segregated subsegment of the broader automobile market with little competitive overlap from other segments of the industry (e.g., luxury sports cars), we score this subfactor based on changes in global market share in that subsegment. Many automobile manufacturers renew their models on a five- to seven-year schedule and attempt to stagger the scheduled renewals of different models, but they may make minor updates in the interim. In cases where consistent market share data are not available, we may estimate the historical changes in global market share.

We may also adjust the subfactor score to differentiate between companies that are experiencing long-term upward or downward trends in market share and companies that are experiencing cyclical or one-time changes that are likely to reverse in the near future. For instance, for a company that had a temporary disruption in production that was subsequently resolved and that is unlikely to recur, we may make an adjustment to reflect the underlying sustainable trend.

Market Position and Product Portfolio

We score this subfactor based on our assessment of the relative market position of the company and the breadth and strength of its product offering. We typically consider a number of aspects of a company's competitive position, with particular emphasis on its footprint in key markets and in the various segments, the strength and profitability of its products, and its strategy and ability to meet future regulatory standards.

Our assessment of a company's ability to offer a broad and diverse range of products within its key geographic markets is typically based on the company's track record as well as our view of its product strategies with respect to consumer trends in these markets and in potential new markets. We also typically assess the company's capacity to successfully carry out those strategies.

While our assessment of this subfactor is largely qualitative, a quantitative measure that can be useful is the product renewal rate, with a higher rate typically indicating a lower risk of obsolescence. In assessing a company's product renewal rate, we generally take into account the potential benefits of a strategically important product launch or an expected acceleration in the pace of product introductions. We also consider the level of profitability of the portfolio and of the different products within the portfolio, with high and even profitability typically resulting in a higher score for this subfactor.

In assessing product breadth and strength, we also consider a company's ability to comply with regulatory standards, mainly for stricter curbs on emissions and tighter enforcement. In particular, we consider a company's record of developing emissions-reducing technologies and alternative-fuel vehicles (AFVs), including the level of innovation and customer acceptance. Given the complexity and variety of emissions regulations across advanced and emerging economies, our assessment also typically focuses on how well articulated and transparent a company's carbon transition strategy is, including the company's ability and willingness to direct sufficient and consistent investments toward meeting future regulatory standards.

Factor: Profitability and Efficiency (20% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBIT Margin.[3]

EBIT Margin

We use the ratio of earnings before interest and taxes (EBIT) to revenue.

Factor: Leverage and Coverage (25% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on four quantitative subfactors: Debt/EBITDA; EBIT/Interest Expense; RCF/Debt; and FCF/Debt.[4]

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

EBIT / Interest Expense

We use the ratio of earnings before interest and taxes (EBIT) to interest expense.

RCF / Debt

We use the ratio of retained cash flow (RCF) to total debt.

FCF / Debt

We use the ratio of free cash flow (FCF) to total debt.

Factor: Financial Policy (15% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

Some companies hold large cash balances in excess of operating and liquidity needs in particular in excess of negative working capital. Where large cash holdings are a reflection of conservative financial policies, this may be an indication of an enduring approach that benefits a company's creditors. On the contrary, when a company maintains cash balances below negative working capital, this may be an indication of more aggressive financial policies.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower

score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[5]

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[6]

Captive finance subsidiaries

Some automobile manufacturers have captive finance subsidiaries that provide inventory financing for dealers and automotive financing and lease financing for retail customers.

Ownership of a captive finance subsidiary can be particularly helpful for an automobile manufacturer during economic downturns because the captive finance subsidiary can be used to provide subsidies to customers, which in turn helps support sales volumes. Beyond this important strategic role, captive finance subsidiaries also can place stress on an automobile manufacturer's financial flexibility. This stress can result from the finance operation's need for supplementary equity capital and liquidity, which are typically formalized in a support agreement. Formal support agreements are typically granted by the automobile manufacturer for the benefit of the finance company and the holders of the finance company's rated debt.

In our assessment of the potential negative effects of a captive finance subsidiary on the creditworthiness of an automobile manufacturer, we typically consider the likelihood and cost of the support that the automobile manufacturer may extend. For details on how we assess the credit impact of a captive finance subsidiary on an automobile manufacturer, see our methodology for rating captive finance subsidiaries of nonfinancial corporations.[7]

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

In this sector, regulatory considerations also play a role in our assessment of a company's profitability and business profile. For instance, changing regulation may require different product offerings, which could affect a company's market position and R&D investments. However, we may assess certain aspects of regulatory considerations outside of the scorecard, for instance when considering the effects on a company's credit quality if regulatory change is swift.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

For example, material changes in accounts payable days on hand can signal a changing dynamic between an automobile manufacturer and its suppliers. Supplier relationships are of particular importance to automobile manufacturers, because many suppliers provide specialized, high-value-added products that may be tailored to a particular brand or model. A shortening of the number of days on hand can signal lower supplier confidence in the automobile manufacturer. A higher number of days on hand could be a sign that the automobile manufacturer can command better credit terms from suppliers, but it can also be a sign that the automobile manufacturer is short of liquidity. This is a particular concern for an automobile manufacturer with low or materially decreasing profitability.

Furthermore, in a downturn, high reliance on trade credit can expose an automobile manufacturer to a more rapid deterioration of working capital.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[8]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Seasonality

Seasonality is an important driver of customer demand and can cause swings in cash balances and working capital positions for companies in this sector. Higher volatility creates less room for errors in meeting customer demand or operational execution.

Cyclical sectors

In a cyclical sector, the scorecard-indicated outcome may be higher than the assigned rating at the top of the economic cycle and lower than the assigned rating at the bottom of the cycle. While using annual or other 12-month-period financials in the scorecard typically provides useful insights into recent or near-term financial results, ratings may also reflect our expectations for the progression of financial results over a longer period that may include a full economic cycle. However, cyclicality itself poses many different types of risks to companies, and cycles do not reverse themselves with predictable regularity. A secular decline or expansion may also affect a cyclical sector. Ratings may incorporate these considerations qualitatively.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4

Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[10]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[11]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[12]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[13] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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8 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

12 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

13 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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REPORT NUMBER 1454190

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